

TIC 257747010

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 2.117424 d (ALIASED, see below)
-> corrected period: 4.234848 d
Epoch (t0) 2798.5052 BTJD
Duration 0.0840 d
Depth 25.379 % (253,790 ppm)
SNR 208.1

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

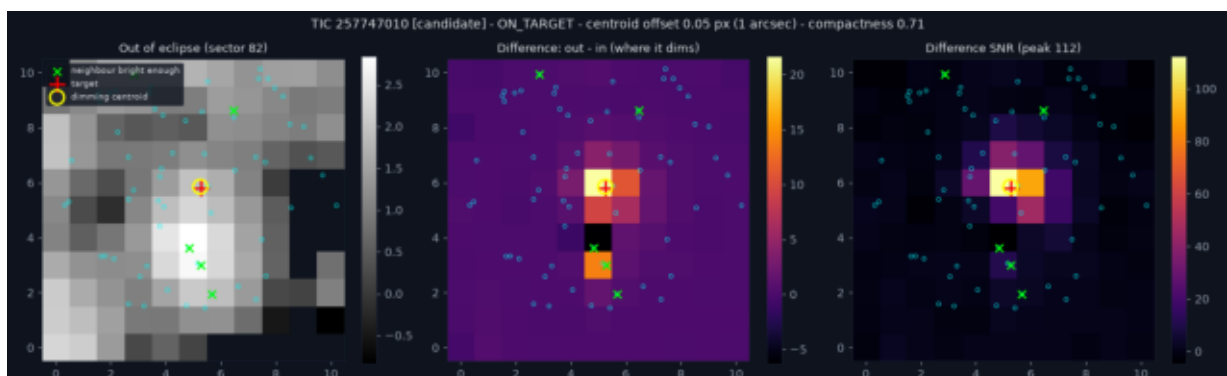
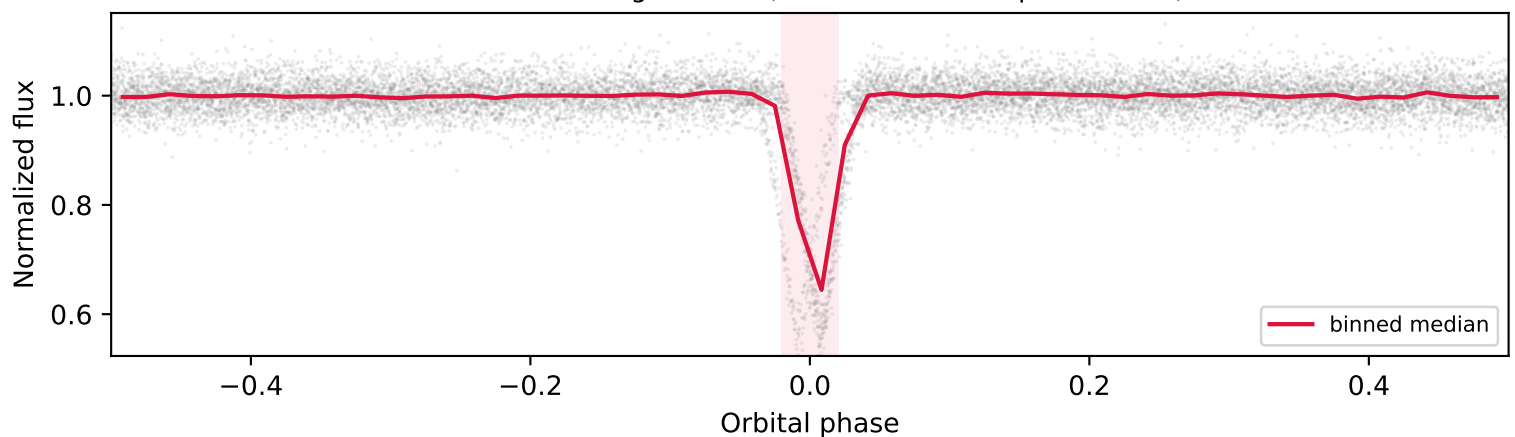
Eclipses observed 23 (gate ≥ 3)
Robust depth ratio 1.11 (gate ≥ 0.50)
Robust depth sigma 181.7 (gate ≥ 7.0)
Odd/even z-score 7.66 (ALIAS FLAGGED)
Secondary eclipse detected, phase 0.04, 5.7 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 82 light-curve dip in this sector: 138.4 sigma t0 refined by 0.0100 phase
Difference-image peak SNR 112.38 (gate ≥ 5.0) Compactness 0.71 (gate ≥ 0.50)
Dimming centroid offset 0.05 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 282944903
2.26 px away, dTmag -2.67 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 7.66 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 4.234848 d, in any write-up.